

Applied Mathematics	Numerical Analysis
Closed-form vs. Algorithmic Solutions	NOTE
Function (Map)	Epigraph of function
Hypograph of function	Injective (1-1)
Surjective (Onto)	Bijjective (Invertible)

• Study of algorithms approximating mathematical analysis • "Calculus done approximately"	Math believed to more directly solve real problems												
Higher degree k kept in Taylor's polynomial leads to better approximation	<table> <tr> <th>Closed-form</th><th>Algorithmic</th></tr> <tr> <td>+ Exact solutions</td><td>+ Broadly applicable</td></tr> <tr> <td>+ Often enable exact theory</td><td>+ Implement well</td></tr> <tr> <td>- Often hard to obtain</td><td>+ Scale well</td></tr> <tr> <td>- Implement poorly</td><td>• Imprecise solutions</td></tr> <tr> <td>- Scale poorly</td><td>• Hard to analyze theoretically</td></tr> </table>	Closed-form	Algorithmic	+ Exact solutions	+ Broadly applicable	+ Often enable exact theory	+ Implement well	- Often hard to obtain	+ Scale well	- Implement poorly	• Imprecise solutions	- Scale poorly	• Hard to analyze theoretically
Closed-form	Algorithmic												
+ Exact solutions	+ Broadly applicable												
+ Often enable exact theory	+ Implement well												
- Often hard to obtain	+ Scale well												
- Implement poorly	• Imprecise solutions												
- Scale poorly	• Hard to analyze theoretically												
For functions with an ordered $\mathcal{Y}$, the epigraph of f is the set $\text{epi}(f) = \{(x, y) \in \mathcal{X} \times \mathcal{Y} : y \geq f(x)\} \subset \mathcal{X} \times \mathcal{Y}$	A rule that assigns every element in $\mathcal{X}$ (the domain) exactly one element in $\mathcal{Y}$ (the codomain) $x \xrightarrow{f} y; y = f(x) \text{ where } x \in \mathcal{X}, y \in \mathcal{Y}$												
f is injective if each $y \in \mathcal{Y}$ is mapped to by AT MOST one x	For functions with an ordered $\mathcal{Y}$, the hypograph of f is the set $\text{hyp}(f) = \{(x, y) \in \mathcal{X} \times \mathcal{Y} : y \leq f(x)\} \subset \mathcal{X} \times \mathcal{Y}$												
f is bijective if it is injective AND surjective	f is surjective if each $y \in \mathcal{Y}$ is mapped to by AT LEAST one x												

Convex Set	Convex function
Concave function	Naming functions
Forms	Quadratic Form
Degree d form number of terms	Homogeneous polynomials (forms)
Univariate Polynomial	Multivariate Polynomial

A function $f : \mathcal{X} \mapsto \mathbb{R}$ is convex if $\text{epi}(f)$ is convex	A set $\mathcal{X}$ is convex if it contains the line segments between any 2 points in it																									
<table><tr><td></td><td>Scalar</td><td>Vector</td><td>Matrix</td><td>Tensor</td></tr><tr><td>Scalar</td><td>Scalar fnctn</td><td>Scalar field</td><td>Scalar field</td><td>Scalar field</td></tr><tr><td>Vector</td><td>V-val fnctn</td><td>Vector field</td><td>V-val fnctn</td><td>V-val fnctn</td></tr><tr><td>Matrix</td><td>M-val fnctn</td><td>Matrix field</td><td>M-val fnctn</td><td>M-val fnctn</td></tr><tr><td>Tensor</td><td>T-val fnctn</td><td>Tensor field</td><td>T-val fnctn</td><td>Tensor field</td></tr></table>		Scalar	Vector	Matrix	Tensor	Scalar	Scalar fnctn	Scalar field	Scalar field	Scalar field	Vector	V-val fnctn	Vector field	V-val fnctn	V-val fnctn	Matrix	M-val fnctn	Matrix field	M-val fnctn	M-val fnctn	Tensor	T-val fnctn	Tensor field	T-val fnctn	Tensor field	A function $f : \mathcal{X} \mapsto \mathbb{R}$ is convex if $\text{epi}(-f)$ is convex
	Scalar	Vector	Matrix	Tensor																						
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$g(\mathbf{x}) = \sum_{i,j=1}^n a_{ij}x_i x_j = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ with $\mathbf{A} \in \mathbb{R}^{n \times n}$ If $\mathbf{A}$ is not symmetric, replace it with the symmetrization of $\mathbf{A}$ $\mathbf{A} = \frac{\mathbf{A} + \mathbf{A}^\top}{2}$	• degree 0 form = constant • degree 1 form = linear form • degree 2 form = quadratic form • $\vdots$																									
A polynomial is homogeneous if the total degree of its terms is the same	Degree d form has $\binom{(n-1)+d}{d}$ terms																									
• $g = \sum_{\mathbf{i} \in \mathcal{I}} \beta_{\mathbf{i}} \mathbf{x}_{\mathbf{i}}$ • $\mathbf{x}_{\mathbf{i}} = \prod_{k=1}^n x_k^{i_k}$ • Total degree of $g = \max_{\mathbf{i} \in \mathcal{I}} \mathbf{1}_n^\top \mathbf{i} = \text{maximum sum of powers of terms}$	• $g = \sum_{\mathbf{i} \in \mathcal{I}} \beta_{\mathbf{i}} \mathbf{x}_{\mathbf{i}}$ • Degree of $g = \max \mathcal{I}$																									

Arity	Linear Transformation
Kronecker Product $A \otimes B$	Vectorization via the Kronecker Product
Vectorization	Why is linearity important?
NOTE	Gradient vs. Derivative
Component Functions	Jacobian Structure of $Df(x_0)$ $f : \mathbb{R}^n \mapsto \mathbb{R}^m$

A function $T : \mathcal{V} \mapsto \mathcal{W}$ between vector spaces is a linear transformation if $\forall \mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $c \in \mathcal{F}$: • $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ • $T(c\mathbf{v}) = cT(\mathbf{v})$ • $T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$	The number of arguments that a function accepts
$\text{vec}(\mathbf{A}\mathbf{B}\mathbf{C}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$	If $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$, their Kronecker product is defined $\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}$ becomes a $mp \times nq$ matrix
Linearity provides the following benefits: • Communicability and Analyticity • Computability • Availability • Mutability	Anything multivariate can be converted to a vector via vectorization Example: $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \implies \text{vec}(\mathbf{A}) = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{bmatrix}$
The gradient is a column vector $\nabla f(x, y) = \begin{bmatrix} f_x(x, y) \\ f_y(x, y) \end{bmatrix}$ The derivative is a row vector $Df(x, y) = [f_x(x, y) \quad f_y(x, y)]$	The gradient is a vector field
$D\mathbf{f}(\mathbf{x}_0) = \begin{bmatrix} \nabla f_1(\mathbf{x}_0)^\top \\ \vdots \\ \nabla f_m(\mathbf{x}_0)^\top \end{bmatrix}$	$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \xrightarrow{f} \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} f_1(x_1, \dots, x_n) \\ \vdots \\ f_n(x_1, \dots, x_n) \end{bmatrix}$

Second Derivative $f : \mathbb{R} \mapsto \mathbb{R}$	Linear Algebra
Derivative Definition via $x - x_0$	Derivative Definition via h
Derivative with Landau	Linearization $f : \mathbb{R} \mapsto \mathbb{R}$
Chain Rule $f : \mathbb{R} \mapsto \mathbb{R}$	Linearization $f : \mathbb{R}^2 \mapsto \mathbb{R}$
Linearization $f : \mathbb{R}^n \mapsto \mathbb{R}$	Chain Rule $f : \mathbb{R}^n \mapsto \mathbb{R}$

The study of vector spaces and the linear transformations between them	$f''(x) = f^{(2)}(x) = \frac{\partial^2 f(x)}{\partial x^2}$
$f : \mathbb{R} \mapsto \mathbb{R}$ is differentiable at a point $x = x_0$ iff $\lim_{h \rightarrow 0} \frac{f(x_0+h)-f(x_0)}{h} \text{ exists}$	$f : \mathbb{R} \mapsto \mathbb{R}$ is differentiable at a point $x = x_0$ iff $\lim_{x \rightarrow x_0} \frac{f(x)-f(x_0)}{x-x_0}$
• $L_{x_0}(x) = f(x_0) + f'(x_0)(x - x_0)$ • The graph is a line in 2d space	• $f(x+h) = f(x) + f'(x)h + o h$ • $f(x) = f(x_0) + f'(x_0)(x - x_0) + o(x - x_0)$
• $L_{x_0,y_0}(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x - x_0) + f_y(x_0,y_0)(y - y_0)$ • The graph is a surface in 3d space	If $g : \mathbb{R} \mapsto \mathbb{R}$ and $h : \mathbb{R} \mapsto \mathbb{R}$ are differentiable, so is $f = h \circ g \text{ and}$ $f'(x) = h'(g(x)) g'(x)$
If $g : \mathbb{R}^n \mapsto \mathbb{R}^m$ and $h : \mathbb{R}^m \mapsto \mathbb{R}$ are differentiable, then $f = h \circ g : \mathbb{R}^n \mapsto \mathbb{R}$ has $\nabla f(\mathbf{x}) = \nabla h(g(\mathbf{x}))^\top Dg(\mathbf{x})$	• $L_{\mathbf{x}_0}(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$ • The graph is an n-dimensional hypersurface in $(n+1)$ dimensional space

Linearization

$$f : \mathbb{R}^n \mapsto \mathbb{R}^m$$

Chain Rule

$$\boldsymbol{f} : \mathbb{R}^n \mapsto \mathbb{R}^m$$

Hessian

$$f : \mathbb{R}^n \mapsto \mathbb{R}$$

Clairaut's Theorem

Differential

Identification Theorems

First Identification Theorem

Second Identification Theorem

NOTE

Derivative of Inverse Matrix

If $\mathbf{g} : \mathbb{R}^n \mapsto \mathbb{R}^p$ and $\mathbf{h} : \mathbb{R}^p \mapsto \mathbb{R}^m$ are differentiable, then $\mathbf{f} = \mathbf{h} \circ \mathbf{g} : \mathbb{R}^n \mapsto \mathbb{R}^m$ has $D\mathbf{f}(\mathbf{x}) = D\mathbf{h}(\mathbf{g}(\mathbf{x})) D\mathbf{g}(\mathbf{x})$	• $L_{\mathbf{x}_0}(\mathbf{x}) = \mathbf{f}(\mathbf{x}_0) + D\mathbf{f}(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$ • The graph is an n-dimensional hypersurface in $(n + m)$ dimensional space
$f(x_i x_j = f_{x_j x_i}) \implies \mathbf{H}(\mathbf{x}) \in \mathbb{R}^{n \times n}$ is symmetric	2nd derivative of scalar field = Jacobian of $\nabla f(\mathbf{x})$ $H(\mathbf{x}) = \frac{\partial^2}{\partial \mathbf{x} \partial \mathbf{x}^\top} f(\mathbf{x}) = [f_{x_i x_j}(\mathbf{x})]$ $\mathbf{H} : \mathbb{R}^n \mapsto \mathbb{R}^{n \times n}$
The means by which derivatives are differentials are linked	$d\mathbf{F}$ of $\mathbf{F}(\mathbf{X})$ = linear part of the change in $\mathbf{F}$ due to small change $d\mathbf{X}$ $d\mathbf{F} := \mathbf{F}(\mathbf{X} + d\mathbf{X}) - \mathbf{F}(\mathbf{X})$ • $d\mathbf{F}$ is always the same size as $\mathbf{F}$ • We cannot “divide by $d\mathbf{X}$” to get a derivative • $d\mathbf{F}$ is a linear function of $d\mathbf{X}$, but may depend nonlinearly on $\mathbf{X}$
• For 2nd derivatives (Hessians) • For $f : \mathbb{R}^n \mapsto \mathbb{R}$ if $d^2 f = (d\mathbf{x})^\top \mathbf{B}(\mathbf{x}) d\mathbf{x}$ then $\mathbf{H}(\mathbf{x}) = \mathbf{B}(\mathbf{x}) + \mathbf{B}(\mathbf{x})^\top$	• For 1st derivatives (Jacobians) • $d\mathbf{f} = \mathbf{A}(\mathbf{x}) d\mathbf{x} \iff D\mathbf{f}(\mathbf{x}) = \mathbf{A}(\mathbf{x})$
If $\mathbf{F}(\mathbf{X}) = \mathbf{X}^{-1}$ $D\mathbf{F}(\mathbf{X}) = -(\mathbf{X}^{-1})^\top \otimes \mathbf{X}^{-1}$	The differential of an object is always its same size. The same cannot be said for the derivative.

Differential Identities: Product Rules	Differential Identities: Trace and Determinant
Differential Identities: Chain Rule	Differential Identities: Basic Arithmetic Rules
Rate of Change	Directional Derivative
Inner Product Space	Inner Product on vector space $\mathcal{V}$
Root Finding	Root Finders/Solvers

• $d \operatorname{tr}(\mathbf{A}\mathbf{X}\mathbf{B}) = \operatorname{tr}(\mathbf{A} (d\mathbf{X}) \mathbf{B})$ • Special case: $d \operatorname{tr}(\mathbf{X}) = \operatorname{tr}(d\mathbf{X})$ • $d \mathbf{X} = \mathbf{X} \operatorname{tr}(\mathbf{X}^{-1}d\mathbf{X})$ • $d \log \mathbf{X} = \operatorname{tr}(\mathbf{X}^{-1}d\mathbf{X})$ • $d\mathbf{X}^{-1} = -\mathbf{X}^{-1} (d\mathbf{X}) \mathbf{X}^{-1}$	• $d(\mathbf{X}\mathbf{Y}) = (d\mathbf{X})\mathbf{Y} + \mathbf{X} (d\mathbf{Y})$ • $d(\mathbf{x}^\top \mathbf{y}) = (d\mathbf{x})^\top \mathbf{y} + \mathbf{x}^\top (d\mathbf{y})$ • $d(\mathbf{x}^\top \mathbf{x}) = 2\mathbf{x}^\top d\mathbf{x}$ • $d(\mathbf{x}^\top \mathbf{A}\mathbf{x}) = \mathbf{x}^\top (\mathbf{A} + \mathbf{A}^\top) d\mathbf{x}$
Assume $\mathbf{A}$ and $\mathbf{B}$ are constant matrices • $d\mathbf{A} = \mathbf{0}$ • $d(\mathbf{A}\mathbf{X}\mathbf{B}) = \mathbf{A} (d\mathbf{X}) \mathbf{B}$ • $d(\mathbf{X} \pm \mathbf{Y}) = d\mathbf{X} \pm d\mathbf{Y}$ • $d(\mathbf{X}^\top) = (d\mathbf{X})^\top$	If $\mathbf{f} = \mathbf{h} \circ \mathbf{g}$ $d\mathbf{f} = \mathbf{A}(\mathbf{g}) d\mathbf{g} = \mathbf{A}(\mathbf{g}) \mathbf{B}(\mathbf{x}) d\mathbf{x}$ • $\mathbf{A}(\mathbf{g}) = D\mathbf{h}(\mathbf{g})$ • $\mathbf{B}(\mathbf{x}) = D\mathbf{g}(\mathbf{x})$
For $f : \mathbb{R}^n \mapsto \mathbb{R}$, the directional derivative of f at $\mathbf{x}_0$ along the vector $\mathbf{v}$ is $\left. \frac{\partial}{\partial t} f(\mathbf{x}_0 + t\mathbf{v}) \right _{t=0} = \left. \frac{\partial}{\partial t} f(\mathbf{c}(t)) \right _{t=0}^\top$ where $\mathbf{c}(t) = \mathbf{x}_0 + t\mathbf{v}$ is a path whose image is a curve in $\mathbb{R}^n$ the rate of change of f at $\mathbf{x}_0$ in the direction of $\mathbf{v}$	$\nabla f(\mathbf{x}_0)^\top \mathbf{v} = \ \nabla f(\mathbf{x}_0)\ _2 \ \mathbf{v}\ _2 \cos \theta$ Maximized when $\cos \theta$ is maximized ie $\theta = 0$
An inner product on a vector space $\mathcal{V}$ over $\mathbb{C}$ is a function $\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \mapsto \mathbb{C}$ such that for $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$ and $a \in \mathbb{C}$ 1. $\langle \mathbf{x} + \mathbf{z}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{z}, \mathbf{y} \rangle$ 2. $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$ 3. Conjugate symmetric: $\langle \mathbf{x}, \mathbf{y} \rangle = \overline{\langle \mathbf{y}, \mathbf{x} \rangle}$ 4. PD : $\langle \mathbf{v}, \mathbf{v} \rangle > 0 \ \forall \text{ nonzero } \mathbf{v} \in \mathcal{V}$	A vector space with an inner product
Algorithms that find roots	Numerically solves $f(x) = 0$ for some function f and input x

NOTE	Sign Function (1d)
NOTE	<code>uniroot()</code>
Multivariate Root Finding	Multivariate Root Finder in R
Multivariate Secant Method (Broyden's Method)	Random Number Generators (RNGs)
Affine Transformations of Normals	Goodness of Fit (GoF) Tests

$\text{sgn}(x) = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \\ 0 & x = 0 \end{cases}$	There is a bisection/grid search hybrid
• R's built-in univariate root finder • numerically solves $f(x) = 0$ for x • essentially a bisection followed by a secant method	Newton's Method need not converge
<code>rootSolve::multiroot()</code>	numerically solves $f(\mathbf{x}) = \mathbf{0}_m$ for $\mathbf{f} : \mathbb{R}^n \mapsto \mathbb{R}^m$
Algorithms used to draws random numbers	Given $\mathbf{x}_n, \mathbf{J}_n$, and $\mathbf{f}(\mathbf{x}_n)$ 1. Solve for Newton-like step: $\mathbf{J}_n \mathbf{h}_n = -\mathbf{f}(\mathbf{x}_n)$ 2. Update iterate: $\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{h}_n$ 3. Evaluate $\mathbf{f}(\mathbf{x}_{n+1})$ 4. Update Jacobian approximation via rank-1 update: $\mathbf{J}_{n+1} = \mathbf{J}_n + \frac{(\mathbf{f}(\mathbf{x}_{n+1}) - \mathbf{f}(\mathbf{x}_n) - \mathbf{J}_n \mathbf{h}_n) \mathbf{h}_n^\top}{\mathbf{h}_n^\top \mathbf{h}_n}$
1. Did my data come from this specific distribution $H_0 : F_X = F_0$ 2. Did my data come from a distribution in this statistical model $H_0 : F_X \in \mathcal{F}$ We usually want to FTR GoF tests. Examples: Pearson's χ^2, Kolmogorov-Smirnov (KS), Shapiro-Wilks	$\mathbf{X} \sim N_p(\mu, \Sigma), \mathbf{A} \in \mathbb{R}^{p^\top \times p}, \mathbf{b} \in \mathbb{R}^{p^\top}$ $\mathbf{A}\mathbf{X} + \mathbf{b} \sim N_{p^\top}(\mathbf{A}\mu + \mathbf{b}, \mathbf{A}\Sigma\mathbf{A}^\top)$

Statistical Model	Dvoretzky-Krefer-Wolfowitz (DKW) inequality
NOTE	ECDF Confidence Bands
Empirical QF (EQF)	What is the basic PRNG in R
Draws vs. Samples	Transformation Theorem
Zero-hypograph of $h(x)$	Auxillary Variable

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F_X(x)$ and let $\hat{F}_n(x)$ be the ECDF. Then for every $\epsilon > 0$, $P\left(\sup_{x \in \mathbb{R}} \left \hat{F}_n(x) - F_X(x) \right > \epsilon\right) \leq 2e^{-2n\epsilon^2}$ Its conservative: it is calibrated for the worst case over all distributions F_X	A set of probability distributions
Solving $2e^{-2n\epsilon^2} = \alpha$ gives $\epsilon_{n,\alpha} = \sqrt{\frac{\log(2/\alpha)}{2n}}$ Since $0 \leq F_X(x) \leq 1$, a 100% simultaneous confidence band is $\max\left\{0, \hat{F}_n(x) - \epsilon_{n,\alpha}\right\} \leq F_X(x) \leq \min\left\{1, \hat{F}_n(x) + \epsilon_{n,\alpha}\right\}$	Know how to prove Chebyshev's Inequality!
<code>runif()</code>	The QF of the empirical distribution • $\hat{Q}_n(p) = \left(\hat{F}_n\right)^{\leftarrow}(p)$ • $\hat{Q}_n(p) \xrightarrow{P} Q_X(p)$
Let $\mathbf{X}$ be a p-dimensional random vector and $\mathbf{g} : \mathbb{R}^p \mapsto \mathbb{R}^n$ be a diffeomorphism. If $\mathbf{X}$ has PDF $f_{\mathbf{X}}(\mathbf{x})$, $\mathbf{Y} = \mathbf{g}(\mathbf{X})$ has PDF $f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{g}^{-1}(\mathbf{y})) \left \frac{\partial}{\partial \mathbf{y}} \mathbf{g}^{-1}(\mathbf{y}) \right $ where $\left \frac{\partial}{\partial \mathbf{y}} \mathbf{g}^{-1}(\mathbf{y}) \right$ is the determinant of the Jacobian matrix of $\mathbf{g}^{-1}$	• To draw = to generate a synthetic observation from • To sample = to draw • A draw = a single synthetic observation from a distribution • A sample = a collection of draws
variable introduced to the problem to help resolve it in some way, but not of interest in the end	$\text{hyp}_0(h) := \{(x, y) : 0 \leq y \leq h(x)\} \subset \mathcal{X} \times \mathcal{Y}$

Monte Carlo Integration (MI)	Grid Search
Pattern Search	Bisection Method
Bisection Method Advantages and Disadvantages	Sign Function (<i>nd</i>)
“Gradient Descent” for solving $f(x) = 0$	Newton’s Method
Secant Method	Multivariate “Gradient Descent”

• Evaluate $f(x)$ on a grid over its domain, look for the rest • Advantages: Easy to implement, gradient free, trivially parallelizable, works well for low dimensions, gives a global sense of $f(x)$ • Disadvantages: Incredibly inefficient, provides imprecise results	Suppose $h : \mathbb{R}^n \mapsto \mathbb{R}$ is an integrable function $I = \int_{\mathbf{A}} h(\mathbf{x}) d\mathbf{x}$
Assumptions: $f(x)$ is continuous and $f(a)f(c) < 0$ 1. Bisect $f(a)$ and $f(c)$ to get $f(b)$ 2. Set $f(b) = f(a)$ if $f(b) < 0$ and $f(b) = f(c)$ if $f(b) > 0$ 3. Repeat until $f(b) < \text{tol}$	1. Start at initial guess x_0 for r 2. Evaluate $f(x_0 \pm \alpha)$ where $\alpha = \text{learning rate}$ 3. If either $f(x_0 \pm \alpha) < f(x_0)$, set $x_1 = x_0 \pm \alpha$, else $\alpha = \alpha/2$ and $x_1 = x_0$ 4. Repeat until $f(x_n) < \text{tol}$ OR a predetermined number of steps has elapsed
$\text{sgn}(\mathbf{x}) = \begin{cases} \frac{\mathbf{x}}{\ \mathbf{x}\ _2} & \mathbf{x} \neq \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{x} = \mathbf{0}_n \end{cases}$	• Advantages: easy to implement, gradient free, much more efficient than grid search • Disadvantages: inefficient compared to gradient methods, doesn't scale to 2^+d
• Requires initial point x_0 to evaluate $f(x)$ and $f'(x)$ • Newton updates: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$	$x_{n+1} = x_n - \alpha \text{sgn}(f(x_n)) \text{sgn}(f'(x_n))$
$x_{n+1} = x_n - \alpha \text{sgn}(f(x_n)) \text{sgn}(\nabla f(x_n))$	Secant updates: $x_{n+1} = \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$

Draws Representation	Law of Large Numbers (LLN)
Central Limit Theorem (CLT)	Multivariate Law of Large Numbers (LLN)
Multivariate Central Limit Theorem (CLT)	Multivariate Delta Method
CDF $F_X(x)$	Quantile Functions $Q_X(p)$
NOTE	Kolmogorov-Smirnov (KS) Test

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F_X(x)$, a distribution with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X) < \infty$ $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu \text{ as } n \rightarrow \infty$	• An infinite number of draws $x_1, x_2, \dots$ • Main Advantage: Everything can be computed almost trivially • Disadvantage: We never have an infinite number of draws
Let $\mathbf{X}_1, \dots, \mathbf{X}_n \stackrel{\text{iid}}{\sim} \mathbf{F}_{\mathbf{X}}(\mathbf{x})$, a distribution with mean $\mu = \mathbb{E}[\mathbf{X}] \in \mathbb{R}^p$ and variance $\Sigma = \text{Var}(\mathbf{X}) \in \mathbb{R}^{p \times p}$ $\bar{\mathbf{X}}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \xrightarrow{a.s.} \mu \text{ as } n \rightarrow \infty$	Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F_X(x)$, a distribution with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X) < \infty$ $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2) \text{ as } n \rightarrow \infty$
If $\mathbf{g} : \mathbb{R}^p \mapsto \mathbb{R}^{p^\top}$ is differentiable at μ, with Jacobian $\mathbf{J} = D\mathbf{g}(\mu) \in \mathbb{R}^{p^\top \times p}$ $\sqrt{n}(\mathbf{g}(\bar{\mathbf{X}}_n) - \mathbf{g}(\mu)) \xrightarrow{d} N_{p^\top}(\mathbf{0}_{p^\top}, \mathbf{J}\Sigma\mathbf{J}^\top)$	Let $\mathbf{X}_1, \dots, \mathbf{X}_n \stackrel{\text{iid}}{\sim} \mathbf{F}_{\mathbf{X}}(\mathbf{x})$, a distribution with mean $\mu = \mathbb{E}[\mathbf{X}] \in \mathbb{R}^p$ and variance $\Sigma = \text{Var}(\mathbf{X}) \in \mathbb{R}^{p \times p}$ $\sqrt{n}(\bar{\mathbf{X}}_n - \mu) \xrightarrow{d} N_p(\mathbf{0}_p, \Sigma) \text{ as } n \rightarrow \infty$
$Q_X(p) : (0, 1) \rightarrow \mathbb{R}$ that, for a given p, solves $P(X \leq x) = p$ • Only nicely exist for univariate continuous data • nondecreasing • left-continuous • $q()$ functions in R	$F_X(x) = P(X \leq x)$ • exists for all distributions • non-decreasing • $F_X(x) \rightarrow 0$ as $x \rightarrow -\infty$; $F_X(x) \rightarrow 1$ as $x \rightarrow \infty$ • right-continuous • $p()$ functions in R
• Tests if two distributions are the same • Built on scaled version of the Fundamental Theorem of Statistics	$\hat{F}_n(\mathbf{x}) \xrightarrow{a.s.} F(\mathbf{x})$

Chebyshev's Inequality	Probability Integral Theorem (PIT)
Inverse Transform Sampling	NOTE
Rejection Sampling	Antidifferentiation vs. Integration
Good Quadrature Rules Aim To:	Simpson's Rule
Gaussian Quadrature	Adaptive Quadrature

Let $X \sim F_X(x)$ be a random variable. Then $Y = g(x) = F_X(x) \sim \text{Unif}(0, 1)$	$P(X - \mu \geq k\sigma) \leq \frac{1}{k^2}$
Know how to prove PIT and Inverse Transform Sampling (See slides 74 and 82 in ‘Random Number Generation’)	Let F_X be a CDF of a random variable with quantile function $Q_X(p)$ and $U \sim \text{Unif}(0, 1)$. Then $Y = Q_X(u) \sim F_X$
• Antidifferentiation asks for a function • Definite integration asks for a number	Produces draws from one the target distribution $f_{\mathbf{X}}(\mathbf{x})$ using draws from the proposal distribution $p_{\mathbf{X}}(\mathbf{x})$ (Derived from Fundamental Theorem of Simulation) • Draw from $p(\mathbf{x})$ • Probabilistically determine if it should be considered from $f(\mathbf{x})$ • Do it over and over
Integrates a quadratic interpolation across pairs of panels ($O(h^4)$)	• use few function evaluations • be exact for low degree polynomials • have stable weights • works well for smooth polynomials
Spends function evaluations where the integrand is hard	• Chooses both nodes and weights intelligently • Effective when f is smooth and function evaluations are expensive

Optimization/Programming	argmin/argmax
Stationary Point	Critical Point
For Convex Optimization	Coordinate Descent
Line Search	Forward Difference
Backwards Difference	Central Difference

the values $\boldsymbol{x}$ you plug in to obtain optima	Maximizing or minimizing a given function $f(\boldsymbol{x})$
Either $\nabla f(\boldsymbol{x}^*) = \mathbf{0}_n$ or the derivative/gradient DNE	A differentiable point where $\nabla f(\boldsymbol{x}^*) = \mathbf{0}_n$
For a given $\boldsymbol{x}_0 \in \mathbb{R}^n$, fix all but the j^{th} coordinate, and optimize only over the j^{th} input exactly or approximately	• Use gradient descent with line search or Newton Methods • stats: Least Squares, Ridge Regression, LASSO, Logistic Regression
$f'(x) \approx \frac{f(x+h)-f(x)}{h}$	Choose a direction $\boldsymbol{d}_k$, then optimize $\phi(\alpha) = f(\boldsymbol{x}_k + \alpha \boldsymbol{d}_k)$
$f'(x) \approx \frac{f(x+h)-f(x-h)}{2h}$	$f'(x) \approx \frac{f(x)-f(x-h)}{h}$

Complex Step Method	Autodiff
Image of A	Preimage of B
Graph of function	Landau Notation
Key Theorem regarding Linear Transformations and Matrices	Multivariate Newton
Monte Carlo Methods	Probability Space

Class of algorithms that compute derivatives	$f'(x) = \frac{\text{Im}(f(x+hi))}{h}$
For $B \subset \mathcal{Y}$, the pre-image of B is $f^{-1}(B) = \{x : f(x) \in B\}$	For $A \subset \mathcal{X}$, the image of A is the set $f(A)$ of $y \in \mathcal{Y}$ mapped to by f from A $f(A) = \{y : \exists x \in A : (x, y) \in f\}$
$f(x)$ is $o(g(x))$ if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0 \iff \lim_{h \rightarrow 0} \frac{f(h)}{g(h)} = 0$	The graph of f is the set $\mathcal{G}(f) = \{(x, f(x)) : x \in \mathcal{X}\} \subset \mathcal{X} \times \mathcal{Y}$ ie the function itself
$\mathbf{x}_{n+1} = \mathbf{x}_n - \mathbf{f}'(\mathbf{x}_n)^{-1} \mathbf{f}(\mathbf{x}_n)$ In practice solve $\mathbf{f}'(\mathbf{x}_n) \mathbf{h}_n = -\mathbf{f}(\mathbf{x}_n)$ and update with $\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{h}_n$	• Every linear transformation between finite-dimensional vector spaces can be represented as a matrix • Matrices are linear transformations once you fix a basis
$(\Omega, \mathcal{B}, P)$ where Ω = sample space, $\mathcal{B}$ = set of events, P = probability space	• Algorithms that rely on repeated random sampling • Built on assumption that we can generate pseudo-observations from a given distribution

Empirical CDF (ECDF)	Fundamental Theorem of Statistics
Probability (PP) Plots	Quantile (QQ) Plots
Box-Muller Transform	Fundamental Theorem of Simulation
Rejection Sampling Scheme	Fundamental Theorem of Calculus
Quadrature Rule	Newton-Cotes

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F_X(x)$. Then $\sup_{x \in \mathbb{R}} \hat{F}_n(x) - F_X(x) \xrightarrow{a.s.} 0 \text{ as } n \rightarrow \infty$ ie the biggest vertical distance between $\hat{F}_n(x)$ and $F_X(x)$ converges to 0.	• The standard basic nonparametric estimator of the CDF • Defined $\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n 1[X_i \leq x]$ – $\hat{F}_n(x)$ = proportion of points $\leq x$ – A random function • ECDF = CDF of discrete distribution that places $1/n$ probability mass at each observation
Scatterplot of $y = X_{(i)}$ versus $x = Q_0(p_i)$	Scatterplot of $y = F_0(X_{(i)})$ versus $x = p_i$
Simulating $\mathbf{X} \sim f_{\mathbf{X}}(\mathbf{x}) \equiv$ simulating $(\mathbf{X}, U) \sim \text{Unif}(\text{hyp}_0(f_{\mathbf{X}}))$ Draws are of the form $(\mathbf{X}, U)$. The variable U is an auxillary variable.	$\mathbf{g}(U) = \mathbf{g}(U_1, U_2) = \sqrt{-2 \log U_1} \begin{bmatrix} \cos(2\pi U_2) \\ \sin(2\pi U_2) \end{bmatrix}$
$\int_a^b f(x) dx = F(b) - F(a)$	Requirement: There exists a known $M > 0 : f(\mathbf{x}) \leq Mp(\mathbf{x}) \ \forall \mathbf{x}$ Algorithm: 1. Sample $\mathbf{x} \sim p(\mathbf{x})$ 2. Sample $U \sim \text{Unif}(0, Mp(\mathbf{x}))$ 3. If $U \leq f(\mathbf{x})$, $\mathbf{X} \sim f(\mathbf{x})$, else repeat
Approximate $f(x)$ by interpolating polynomial on equally spaced nodes, then integrate that polynomial exactly ($O(h^2)$ in general)	Any apporximation of the form $I = \int_a^b f(x) dx \approx \sum_{k=1}^m w_k f(x_k) = \hat{I} \text{ where } x_k = \text{nodes} \text{ and } w_k = \text{weights}$ $\hat{I}$ = linear combination of the points, $\hat{I} = \mathbf{w}^\top \mathbf{f}$

<code>integrate()</code>	Optimization Problem
NOTE	Types of Optimization Problems
NOTE	Forward AD
Forward AD (Continued)	

$\min_{\boldsymbol{x} \in \mathcal{R}} f(\boldsymbol{x}) \text{ or } \max_{\boldsymbol{x} \in \mathcal{R}} f(\boldsymbol{x})$ $f(\boldsymbol{x}) = \text{objective function, mins/maxes are optima,}$ $\mathcal{R} = \text{feasibility region}$	• R's built in function for integration • does 1d adaptive quadrature
1. Convex optimization: $f(\boldsymbol{x})$ is a convex function; $\mathcal{R}$ is a convex set 2. Linear programming: $f(\boldsymbol{x})$ is linear; $\mathcal{R}$ is a polytope 3. Combinatorial optimization: $\mathcal{R}$ is a discrete set (usually enormous)	Mins and maxes interconvert $\min_{\boldsymbol{x} \in \mathcal{R}} f(\boldsymbol{x}) = - \max_{\boldsymbol{x} \in \mathcal{R}} f(\boldsymbol{x})$
$f'(x) = \text{Du} (f (x + 1\epsilon))$	Coordinate/line methods seek to reduce multivariate optimization problems $f : \mathbb{R}^n \mapsto \mathbb{R}$ to univariate optimization problems $f : \mathbb{R} \mapsto \mathbb{R}$
	Let $f(x) = h (g (x))$ then $(x 1\epsilon) \xrightarrow{g} (g(x), g'(x)) \xrightarrow{h} (h (g(x)), h' (g(x)) g'(x))$